



TXW80X Certification Test Guide



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Table of Contents TXW80X Certification Test Guide..... 1 1. Overview..... 1 2. Certification testing..... 1 2.1. Test Overview..... 1 2.2. Test commands.... 2 2.2.1. General TX modulated signal test steps.....2 2.2.2. 20M bandwidth TX modulated signal test.. 2 2.2.3. 10M bandwidth TX modulated signal test.....2 2.2.4. 5M bandwidth TX modulated signal test.....2 2.3. Frequency point list..... 3 2.4. Precautions..... 3				
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1 Overview

This article introduces the solution certification testing method to help and guide you in conducting Wi-Fi 4 EMI testing, which is usually applied to SRRC, FCC, CE, CCC and other certification tests, this article takes the FCC test as an example.

This document is mainly applicable to the following engineers:

• Technical support engineer

• Solution software development engineer

Product scope applicable to this document:

model	encapsulation	Package
TXW806	QFN56/48/40/36/32/24	
TXW803	QFN32	
TXW802	QFN32	
TXW801	QFN24	

2. Certification testing

2.1. Test Overview

Test purpose: Test the TX EMI performance of Wi-Fi 4

Modulation technology: • CCK

• DSSS

• BPSK QPSK 16-QAM 64-QAM for OFDM

Operating frequency: 2412~2462Mhz for 802.11b/g/n

Test method: serial port AT command set, the default serial port baud rate is 115200, please do not use an isolated serial port

2.2. Test commands

2.2.1. 20M bandwidth TX modulated signal test

AT+BSS_BW=20 //Set BSS_BW to 20M

AT+TX_FRM_TYPE=0 //Set the sending format. All sending formats support 20M bandwidth

2.2.2. Universal TX modulated signal test steps

Test command sequence (taking 20M bandwidth as an example, other bandwidth tests need to modify the

settings of steps c and d): a) AT+TEST_START=1 //Enter test

mode b) AT+LO_FREQ=2412 //Set frequency point (MHZ) , please refer to the frequency point list chapter for details. c) AT+BSS_BW=20 //Set BSS_BW to 20M

d) AT+TX_FRM_TYPE=0 //Set the sending format, taking 0:802.11b as an example, the specific correspondence is as follows:

802.11b:

0: Represents sending DSSS/CCK LONG format. Support MCS 0~3

1: Represents sending DSSS/CCK SHORT format. Support MCS 1~3

802.11g: (OFDM) 2:

Represents sending NON-HT format. Support MCS 0~7

802.11n: (OFDM) 3:

Represents sending HT-MF format. Support MCS 0~7

4: Represents sending HT-GF format. Support MCS 0~7

e) AT+TX_MCS=0 //Set the MCS of TX, taking MCS0 as an example, the MCS corresponding to each packet format is

as above. f) AT+TX_START=1 //Enable TX

2.2.3. 10M bandwidth TX modulated signal test

AT+BSS_BW=10 //Set BSS_BW to 10M

AT+TX_FRM_TYPE=2 //Set the sending format, note that only 802.11g supports 10M bandwidth

2.2.4. 5M bandwidth TX modulated signal test

AT+BSS_BW=5 //Set BSS_BW to 5M

AT+TX_FRM_TYPE=2 //Set the sending format, note that only 802.11g supports 5M bandwidth

2.3. Frequency list

The TXW80X working channel number and channel center frequency are as shown in the following table:

Channel	Frequency/MHz	Channel	Frequency/MHz
1	2412	7	2442
2	2417	8	2447
3	2422	9	2452
4	2427	10	2457
5	2432	11	2462
6	2437		

appendix:

Table 2-3-1 Applicable frequency bands in various countries (2.4G)

信道	频率 (MHz)	中国	美国、加拿大	欧洲	日本	澳大利亚	委内瑞拉	以色列
1	2412	是	是	是	是	是	是	否
2	2417	是	是	是	是	是	是	否
3	2422	是	是	是	是	是	是	是
4	2427	是	是	是	是	是	是	是
5	2432	是	是	是	是	是	是	是
6	2437	是	是	是	是	是	是	是
7	2442	是	是	是	是	是	是	是
8	2447	是	是	是	是	是	是	是
9	2452	是	是	是	是	是	是	是
10	2457	是	是	是	是	是	是	否
11	2462	是	是	是	是	是	是	否
12	2467	是	否	是	是	是	是	否
13	2472	是	否	是	是	是	是	否
14	2484	否	否	否	802.11b	否	否	否

The 2.4G frequency used in the United States is 2412~2462MHz (Chan1~Chan11)

2.4. Precautions

Please refer to the EMC laboratory's EMC testing precautions. For example: According to the EMC tested by the EMC laboratory of Zhuhai Beijing Normal University

Test reference precautions, the radiation spurious test is conducted in two parts:

- 30MHz~1GHz: Just test according to the normal configuration.
- 1GHz~25GHz: When testing this item, you need to pay attention to the fact that the main tone is close to 2.4GHz and the laboratory horn antenna

The preamp receiving signal on (1GHz~6GHz) is too large and saturated. The nonlinearity of the instrument makes the measured harmonic energy much larger.

For actual reception, it is necessary to add a band stop filter (2.4 frequency band) before the preamp to attenuate the main tone energy.